

to solve issues in this space. Canada-based Kira Systems uses machine learning (ML) for contract analysis. UK's company Tessian employs AI to secure confidential data and emails for law firms.

In India, researchers from IIT Kharagpur have recently developed an AI-powered method to automate the reading of legal case judgements, case law analysis and enhance legal search across several domains. Deep neural models enable understanding the rhetorical roles of sentences or jargons in a judgement when adequate data is available, and hence aid in organising legal documents.

Blockchain finds application in email encryption, verifying process and securing evidence, like in financial transactions and many other purposes. For example, Legaler's blockchain and developer tools provide infrastructure to build decentralised applications for legal services. In 2017, Global Legal Blockchain Consortium was formed to drive the standardisation of blockchain technology in the legal sector. It has already surpassed 150-member mark, which includes law firms, software companies, and universities, among others.

According to a 2019 report titled LawTech Adoption Research by tech

analysts TechMarketView, the number of lawtech companies has grown over the past few years, but the adoption rate is not that high among legal practitioners. One of the major reasons noted behind this is the partnership model in which spending is done from the partners' profit pool.

To change this scenario, the US states like North Carolina and Florida have already mandated technology training for CLE (continuing legal education) credits. Pressure from clients for cheaper offerings is also pushing law firms to move to cloud computing and other tech solutions.

—Ayushee Sharma

## 2 AIR PURIFICATION FOR SUSTAINABLE LIVING

**With demand for air-purifiers increasing almost all over India, solar-powered and algae-based air-purifiers, which do not need a regular supply of electricity, are also coming to the market**

A majority of the population caught in the grip of acute as well as chronic respiratory ailments like asthma, especially in urban cities today, has air pollution to blame. And this is just one of the devastating impacts caused by air pollutants resulting from vehicular emissions, exhaust from industries, agricultural activities, and so on. Pollution is pushing every sector to switch from unsustainable methods of development to renewable-energy-based solutions.

To serve a large population, inexpensive yet highly efficient air-purification systems that can cover a large effective area in public places like bus stands, traffic signals, and hospitals are required. In China, such a solar-based air-purification tower approximately a hundred metres high was built to reduce smog levels in Xian in Shaanxi province as an experiment a few years ago. A system of greenhouses around the base of the tower traps the polluted air where it gets heated by solar energy. The hot air thus produced rises through the tower and passes through multiple layers of



(Credit: [www.intelligentliving.co](http://www.intelligentliving.co))

cleaning filters.

Growing awareness has made companies realise the importance of small-scale air-purifiers for use in homes, cars, offices and other commercial areas, too. These air-purifiers perform the same function as huge air-purification towers, that is, they can be used to get rid of dust particles, dust mites, smoke particles, vehicle emissions, and gaseous pollutants, among others, but their effective area is small. Air-purifiers rely on multiple or compound filters to clean air by removing suspended particulates through

mechanical separation.

Making air-purifiers is not enough; care must be taken to ensure there are no hazards that may harm the environment or users' health. For example, using an ozone air-purifier may not be a good solution; ozone can be dangerous for human health because it can cause breathing issues and, in severe cases, lung damage, according to the Environmental Protection Agency.

Demand for eco-friendly, economical, energy-saving purifiers has led to the rise of solar purifiers. In these purifiers, the purifier is usually powered through electrical mains but solar energy is available as an alternative, which can be highly beneficial in remote households that are off the electrical grid. This also ensures lower electrical consumption and hence lower operational cost.

Solar-powered air-purifier products are highly popular in places like the United States, China, South Korea, and the Netherlands. Portable solar purifiers for cars from manufacturers like Airnasa and Olansi are the most common. In terms of technology used,